FEDERAL UNIVERSITY OF AMAZONAS
COMPUTING INSTITUTE - IComp
SPHERE RESEARCH GROUP

INFORMED CONSENT FORM

Dear Sir/Madam,

This Informed Consent Form is used for any participation in research studies for the investigation of the adoption and use of Software Engineering (SE) and Testing practices in the development of Artificial Intelligence and Machine Learning (AI/ML) models, as well as in the integration of AI/ML models into software systems. Your participation in this or any study is entirely voluntary. You are free to decline participation or stop answering the survey at any time, without any penalty or negative consequence.

Research Title: *SE4AI: Adoption of Software Engineering Practices in AI/ML Development*

The researchers responsible for this study are available to provide further information and answer any questions you may have.

Principal Investigator:
Felipe Sonntag Manzoni, M.Sc.
Federal University of Amazonas (UFAM) – Institute of Computing
Av. Rodrigo Otávio, No. 6,200, North Sector – Manaus, Amazonas, Brazil – ZIP Code: 69077-000

Supervising Researchers:
Ana Carolina Oran, Ph.D.
Leonardo Carneiro Marques, Ph.D.
Federal University of Amazonas (UFAM) – Institute of Computing
Same address as above

1) General Objective
The general objective of this research is:

1. To investigate the current adoption of Software Engineering and Testing principles and practices in AI/ML development contexts; and
2. To support the design of a structured development process for AI/ML model development and AI/ML-enabled software systems, grounded in evidence from academic and industrial practice.

2) Specific Objectives
The specific objectives of this study are:

a) To identify and assess the adoption of traditional and AI/ML-tailored Software Engineering principles and practices in academic and industrial contexts;
b) To identify and assess the adoption of traditional and AI/ML-tailored Testing

principles and practices in academic and industrial contexts;
c) To examine how quality-related practices are currently incorporated into AI/ML development processes;
d) To analyze existing development procedures and practices adopted in academic and industrial scenarios;
e) To support the proposal of a more structured and quality-oriented development process for AI/ML-enabled software systems.

3) Risks
All research involving human participants involves some level of risk. However, the risks associated with this study are considered minimal.

This study consists exclusively of answering an online survey about Software Engineering and Testing practices used in AI/ML development. The survey does not request directly identifying personal information, nor does it ask sensitive questions about participants' private lives, health, beliefs, or other personal matters unrelated to the research objectives.

Possible risks include mild discomfort, inconvenience, or fatigue while answering the survey. To reduce these risks:

a) Participation is entirely voluntary;
b) You may stop answering the survey at any time, without penalty;
c) You may choose not to answer any question you do not wish to answer;
d) No directly identifying personal information will be requested in the survey;
e) Respondents can stop the survey and return later without penalty from where they stopped;
f) If respondents drop the survey at any point and do not finish it in 24 hours, they will be remembered 1 time only to go back and finish it.

To support analysis of diversity and geographic distribution of responses, the survey may include a question about the participant's country or region of work/study. This information will be used exclusively for research analysis and reporting in aggregated form, without personal identification.

Participation in this survey does not involve any expected financial cost to you, however, if you believe that participation in this study will result in any expense or harm, please contact the research team using the contact information provided in this form before taking the survey. Any applicable assistance, clarification, or compensation procedures will be handled in accordance with the ethical requirements governing the study.

4) Benefits
This study is expected to contribute to a better understanding of how Software Engineering and Testing practices are currently adopted in AI/ML development in both academic and industrial settings.

The expected benefits of the research include:

a) Generating empirical evidence on the adoption of SE and Testing practices in AI/ML development;

b) Identifying challenges, gaps, and opportunities for improvement in current AI/ML development processes;
c) Supporting the future design of a more structured and quality-oriented development process for AI/ML models and AI/ML-enabled software systems.

This study may not provide direct personal benefit to participants. However, its findings may contribute to future research and practice in Software Engineering for AI/ML systems.

5) Procedures
If you agree to participate, you will be asked to complete an online survey questionnaire containing questions about Software Engineering and Testing principles and practices related to the development of AI/ML models and AI/ML-enabled software systems.

The survey responses will be collected and stored only for the time necessary to conduct data organization, cleaning, analysis, and reporting of the research results. Any information collected about participant location, such as country or region, will be used only for analysis of geographic diversity and representation in the study.

The results of the study may be published in academic papers, reports, dissertations, theses, or presentations. In all such cases, the data will be presented in aggregated or anonymized form.

6) Handling of Possible Risks and Discomfort
Your participation consists solely of answering the survey questionnaire, which is expected to take approximately 20 to 40 minutes.

The data collected in this study will be used strictly for research purposes. They will not be used for professional evaluation, employment-related decisions, academic grading, or any form of personal assessment.

You may interrupt your participation and exit the survey at any time, without any penalty or loss of rights.

7) Costs and Compensation
Participation in this survey does not involve any expected financial cost to you, and no payment or financial reward will be provided for participation.

If any exceptional expense directly related to participation in this study occurs, you may contact the research team for clarification and appropriate handling, in accordance with the ethical procedures applicable to the study.

8) Confidentiality
All information collected in this survey will be treated confidentially and used exclusively for academic and scientific purposes.

The survey does not request your name, the name of your organization, or other directly identifying personal information. The research team will make every reasonable effort to preserve confidentiality and to present results only in aggregated or anonymized form.

9) Voluntary Participation
Your participation in this study is voluntary. You have the right not to participate, to stop participating at any time, and to refuse to answer any question, without any penalty or negative consequence.

To participate in this study, you must be 18 years of age or older.

10) Consent Statement
By agreeing to participate, you confirm that:

- You have read, or had read to you, the information contained in this Informed Consent Form;
- You have understood the purpose and procedures of the study;
- You understand that your participation is voluntary;
- You understand that you may stop participating at any time, without penalty;
- You are 18 years of age or older;
- You freely and voluntarily agree to participate in this study.

If you have any questions about this research, you may contact:

Felipe Sonntag Manzoni
Email: *fsm2@icomp.ufam.edu.br*

Questions or concerns regarding the ethical aspects of this research may be directed to:

Research Ethics Committee of the Federal University of Amazonas (CEP/UFAM)
Escola de Enfermagem de Manaus – Room 07 – Rua Teresina, 495 – Adrianópolis, Manaus – AM, Brazil
Telephone: +55 (92) 3305-1181, extension 2004
Email: cep.ufam@gmail.com

This research has been submitted to and reviewed through the appropriate ethical review procedures of the Federal University of Amazonas via Plataforma Brasil.